

Search Engine Ranking Game Teacher Notes

TOK Cognitive Science Activity Bank • Knowledge and Technology • 25-30 min

Category	Knowledge and Technology	Time	25-30 min
Difficulty	Medium	ID	search-engine-ranking-game

Big idea: Visibility is not the same as reliability.

Overview

Students investigate how ranking affects what becomes visible as knowledge.

Student Prompt

Rank the six search results from most reliable to least reliable. Then identify which ranking signal influenced you most: expertise, popularity, freshness, sponsorship, relevance, or emotional appeal.

Concrete Stimulus Packet

Result 1: Ad · PureGreen Cleanse Co.	"7-Day Celery Juice Liver Detox — feel lighter fast."	Sells detox juice packs. No author listed. Ranking signal: Sponsorship.
Result 2: WellnessNow Blog	"I drank celery juice every morning. Here's what changed."	Personal story, dramatic before/after claims, no medical sources. Ranking signal: Emotional appeal.
Result 3: National Liver Foundation — Patient Guide	"Your liver already removes toxins from your blood."	Explains liver function; reviewed by medical advisers. Ranking signal: Expertise.
Result 4: HealthLine Daily	"Celery juice: benefits, myths, and what evidence says."	Notes hydration/nutrients but says detox claims are unsupported. Ranking signal: Relevance.
Result 5: TikTok Clip: @CleanBodyGuru	"Doctors don't want you to know this detox trick."	2.1M views, no evidence provided. Ranking signal: Popularity.
Result 6: Research Summary: Journal of Nutrition Education	"Vegetable intake and health outcomes."	Peer-reviewed; does not test celery juice detox claims directly. Ranking signal: Academic credibility, indirect relevance.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	Which result would you click first, and why?	Look for whether students choose visibility, credibility, relevance, or emotion.
2	Which result is most reliable for the query?	Likely result 3 or 4; result 6 is credible but indirect.

#	Question for students	Teacher key / cue
3	Which result might a popularity-based ranking put first?	Result 5: TikTok Clip @CleanBodyGuru.
4	Which result might a sponsorship-based ranking put first?	Result 1: PureGreen Cleanse Co. advertisement.
5	Which result best shows expertise?	Result 3: National Liver Foundation patient guide.
6	What does this show about visibility and reliability?	A ranking system can make a weak result visible before a strong one.

Teacher Key / Reveal

A search engine could rank result 5 highest by popularity, result 1 highest by sponsorship, result 4 highest by relevance, and result 3 highest by expertise. Visibility is not the same as reliability.

- Most medically reliable: Result 3, then Result 4.
- Most popular-looking: Result 5.
- Most commercially promoted: Result 1.
- Most academically credible but indirect: Result 6.
- Least reliable for knowledge: Result 5 or Result 2, depending on criteria.

Setup Checklist

- Prepare a starter stimulus using: Mock search results ranked by popularity, expertise, freshness, sponsorship, or relevance.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Ready-to-Use Examples

- Mock search results ranked by popularity, expertise, freshness, sponsorship, or relevance.
- A query about a health claim with credible, vague, commercial, and opinion sources.
- Two ranking rules that produce different 'top answers' for the same searcher.

Suggested Timing

Stage	Teacher move	Watch for
1	Present a question such as "Is homework effective?"	Assumptions, confidence shifts, evidence claims, and language choices.
2	Students rank which results they would click first.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Reveal hidden information: sponsored result, old source, expert review, blog post, advocacy group and research article.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Students redesign the search page to make reliability easier to judge.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Write your first judgement before we discuss it; the first answer is useful evidence.

- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Common Misconceptions

- Students may treat a tool's output as neutral because it feels automatic.
- Students may focus only on user error and miss design incentives or hidden rules.

Assessment Look-Fors

- Identifies the rule, design choice, or incentive shaping knowledge.
- Distinguishes access to information from justified knowledge.

Differentiation and Adaptations

- Short lesson: use only the first example, one response grid, and one debrief question.
- Long lesson: add a second round where students revise the method and compare outcomes.
- Low-tech option: run with printed cards, sticky notes, and a board tally.

Extension Tasks

- Design a second version of the activity that tests the same knowledge issue in another AOK.
- Find a real-world example where the same knowledge problem matters outside the classroom.
- Write a 150-word TOK exhibition-style commentary using the activity as the object context.

Related Activities

- Algorithmic Feed Simulation
- AI Hallucination Courtroom
- Deepfake Detection Challenge